
Summary

Senior software engineer and technical lead with 20+ years across embedded systems, cellular/mobile security, and vulnerability research. Leads 5–6 person research teams end-to-end: technical direction, sprint planning, direct technical contributions, and customer-facing delivery. Deep hands-on background in full-stack development (Python, Postgres, JavaScript/TypeScript), reverse engineering (ARM, C/C++, CAN bus).

Employment History

Nightwing

2019 – Present

Technical Lead — Led 5–6 person research teams across multi-year programs. Owned technical direction, sprint planning, and direct technical contributions; delivered regular status to the customer; provided performance feedback for all team members and owned team composition.

- **2019 – 2021 — Firmware CVE tracking effort.** Lead developer for a web crawler that collected firmware images across ~12 vendors, building a ~2 TB corpus used to track CVEs across generations of firmware. (Python, Postgres)
- **2021 – 2024 — Industrial robotics vulnerability research.** Led a team reverse-engineering compiled C/C++ industrial robot systems, including live instrumentation of running systems via the CAN bus.
- **2024 – Present — Physical access-control vulnerability research.** Leading reverse engineering and vulnerability research of building security systems — HID readers, NVRs, and IP cameras. (Python)

Independent Contractor

2016 – 2019

- **Raytheon Intelligence Information Services**
 - Developed a software build system and mirroring infrastructure for an air-gapped network.
- **Facilities Survey** — Lead developer for the CMS View and Space Manager hospital facilities management product suite (JavaScript, TypeScript, Python backend on Microsoft Azure). Multiyear effort spanning multiple disciplines.
 - fsiservices.com/cms-view
 - fsiservices.com/cms-space-manager
- **LandEconics** — Developed a cost projection estimation system for landscape architects.

Endgame

2011 – 2016

Director, Mobile Access

- Led a three-engineer team developing repeatable, isolated, over-the-air black-box test procedures for an unsupported third-party smartphone app, identifying and characterizing bugs across multiple software versions.
- Led a three-person, six-month sub-API-level study of smartphone sandbox limitations across iOS, Android 4.2, Symbian, WP8, and BlackBerry 7.
- Internal research projects repurposing consumer electronics hardware; developed an Android wireless sniffing tool including the front-end implementation.

Stack: ARM architecture, Java, Python, RealView ARM Tools, IDA Pro. *Platforms:* Nokia/Symbian, Android, Windows Phone 8, BlackBerry 7. *Test equipment:* Anritsu, ip.access Picocell, OpenBTS. *Waveforms:* 802.11, GSM, WCDMA/UMTS, CDMA.

Earlier roles (2006 – 2011)

details on request

Additional work history (Raytheon S.I. Government Solutions; Harris Corporation) omitted for space. Full details available on request.

Education

Auburn University — Master of Software Engineering, August 2005

Baylor University — Bachelor of Science (Computer Science), December 2002

Talks & Publications

Malware Technical Exchange Meeting 2015

Exploitation of Mobile OEM Software

DOJ National Security Division's Cyber Awareness Series 2015

Mobile Security Backgrounder

Towards an Indoor Location System Using RF Signal Strength in IEEE 802.11 Networks

Adam Harder, Lanlan Song, Yu Wang

International Conference on Information Technology Coding and Computing, ITCC 2005, Sponsored by IEEE Computer Society, April 4–6, 2005, Las Vegas, NV